TABLEAU – WRITE UP

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ISM 6361 - Data Visualization

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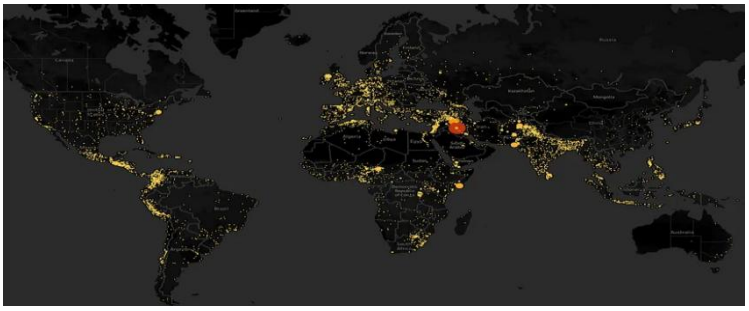
Global Terrorism Database



DATA SET SOURCE: KAGGLE https://www.kaggle.com/datasets/START-UMD/gtd

HYPERLINK FOR MY PROJECT - TABLEAU PUBLIC SITE

https://public.tableau.com/app/profile/tejashwini.saravanan/viz/TheGlobalTerrorsim/TheGlobalTerrorism

BACKGROUND

The world is suffering and bleeding day by day. Too many people are dying among both civilians and soldiers. Words are not needed when you know they will not change this horrifying scenario. How can we fight terrorism, then? The answer is participation, the answer is comprehension. The world needs to stay logical, to not fill its heart with anger, because anger carries blindness.

It is evident that there has been enormous growth in terrorist attacks in recent years. As Data Analytics students, we have the tools of data and visualization with us to answer some of the questions which can help to find out trends, patterns, and some interesting insights from this dataset. This can be our contribution to tackling the horrors of global terrorism. Using this data, we can also predict the terrorist attack using machine learning which will help to reduce the feeling of uncertainty as how the terrorist went about planning a major attack undetected, was the terror act an isolated instance or the first of a series, and who were the responsible groups behind such heinous crimes.

OBJECTIVE OF ANALYSIS AND STUDY

The GLOBAL TERRORISM INDEX defines **terrorism** as "the threatened or actual use of illegal force and violence by a non-state actor to attain a political, economic, religious, or social goal through fear, coercion, or intimidation.

The objectives of this project are:

· To study and analyse the different terrorism acts that have taken place in various parts of the world, at different times.

· The various causes behind terrorism acts

· Different types of terrorism acts

· Categories of terrorism acts

· Distribution of terrorism acts

· Statistical conclusions

About the dataset

The Global Terrorism Database (GTD) is an open-source database including information on domestic and international terrorist attacks around the world from 1970 through 2017, and now includes more than 181,691 cases. For each event, information is available on the date and location of the incident, the

weapons used and nature of the target, the number of casualties, and–when identifiable–the group or individual responsible. There are 18,1691 records and 135 columns including date, time, location, number of hostages, killed, wounded, if there was a ransom, the outcome, if there was a suicide attack, claims, weapons used. Unit of analysis: Attack. The number of variables is > 100 on location, tactics, perpetrators, targets, and outcomes.

Characteristics of the GTD

· Contains information on over 181,691 terrorist attacks

· Currently the most comprehensive unclassified database on terrorist attacks in the world

· Includes information on more than 91,000 bombings, 20,000 assassinations, and 13,000 kidnappings and hostage events since 1970

· Includes information on at least 45 variables for each case, with more recent incidents including information on more than 120 variables

· More than 4,000,000 news articles and 25,000 news sources were reviewed to collect incident data from 1998 to 2017 alone

attacktype	attacktype	attacktype	attacktype	attac	atti	targtype1	targtype1_txt	targs	targsubtype1_txt	corp1	target1	natlty1	natlty1_txt
1	Assassination					14	Private Citizens & Property	68	Named Civilian		Julio Guzman	58	Dominican Republic
6	Hostage Taking (Kidnapping)					7	Government (Diplomatic)	45	Diplomatic Personnel (outside of embass	Belgian Ambassador Daught	Nadine Chaval, daughter	21	Belgium
1	Assassination					10	Journalists & Media	54	Radio Journalist/Staff/Facility	Voice of America	Employee	217	United States
3	Bombing/Explosion					7	Government (Diplomatic)	46	Embassy/Consulate		U.S. Embassy	217	United States
7	Facility/Infrastructure Attack					7	Government (Diplomatic)	46	Embassy/Consulate		U.S. Consulate	217	United States
2	Armed Assault					3	Police	22	Police Building (headquarters, station, sc	Cairo Police Department	Cairo Police Headquarters	217	United States
1	Assassination					3	Police	25	Police Security Forces/Officers	Uruguayan Police	Juan Maria de Lucah/Chief of Directorate	218	Uruguay
3	Bombing/Explosion					21	Utilities	107	Electricity	Pacific Gas & Electric Compa	Edes Substation	217	United States
7	Facility/Infrastructure Attack					4	Military	28	Military Recruiting Station/Academy	R.O.T.C.	R.O.T.C. offices at University of Wiscons	217	United States
7	Facility/Infrastructure Attack					2	Government (General)	21	Government Building/Facility/Office	Selective Service	Selective Service Headquarters in Madisc	217	United States
3	Bombing/Explosion					4	Military	27	Military Barracks/Base/Headquarters/Checkpost		Badger Army ammo depot.	217	United States
7	Facility/Infrastructure Attack					4	Military	28	Military Recruiting Station/Academy	Army Recruiting Station	Army Recruiting Station, Denver Colorad	217	United States
4	Hijacking					6	Airports & Aircraft	42	Aircraft (not at an airport)	Trans World Airline	Flight 802 Boeing 707	217	United States
7	Facility/Infrastructure Attack					2	Government (General)	21	Government Building/Facility/Office	U.S. Government housing	Packard Properties building of Detroit Mi	217	United States
7	Facility/Infrastructure Attack					1	Business	7	Retail/Grocery/Bakery	American owned business in	Baker's Store	217	United States
3	Bombing/Explosion					2	Government (General)				Jurists Ball (Palais au Frankturm)	75	Germany
9	Unknown					4	Military	34	Military Personnel (soldiers, troops, offic	U.S. Army	Soldier	217	United States
3	Bombing/Explosion					8	Educational Institution	49	School/University/Educational Building	High School	James Madison High School	217	United States
3	Bombing/Explosion					1	Business	4	Multinational Corporation	General Electric	General Electric factory in Rio Grande Pu	217	United States
7	Facility/Infrastructure Attack					1	Business	7	Retail/Grocery/Bakery	Fuson's Department Store	Fuson's Department Store, Seattle Washi	217	United States
7	Facility/Infrastructure Attack					3	Police	22	Police Building (headquarters, station, sc	Police Department	Champaign Police Department	217	United States
2	Armed Assault					8	Educational Institution	49	School/University/Educational Building		Secondary School, Chemistry Lab	218	Uruguay

GLOBAL TERRORISM DATABASE FROM KAGGLE

Visualizations ideas

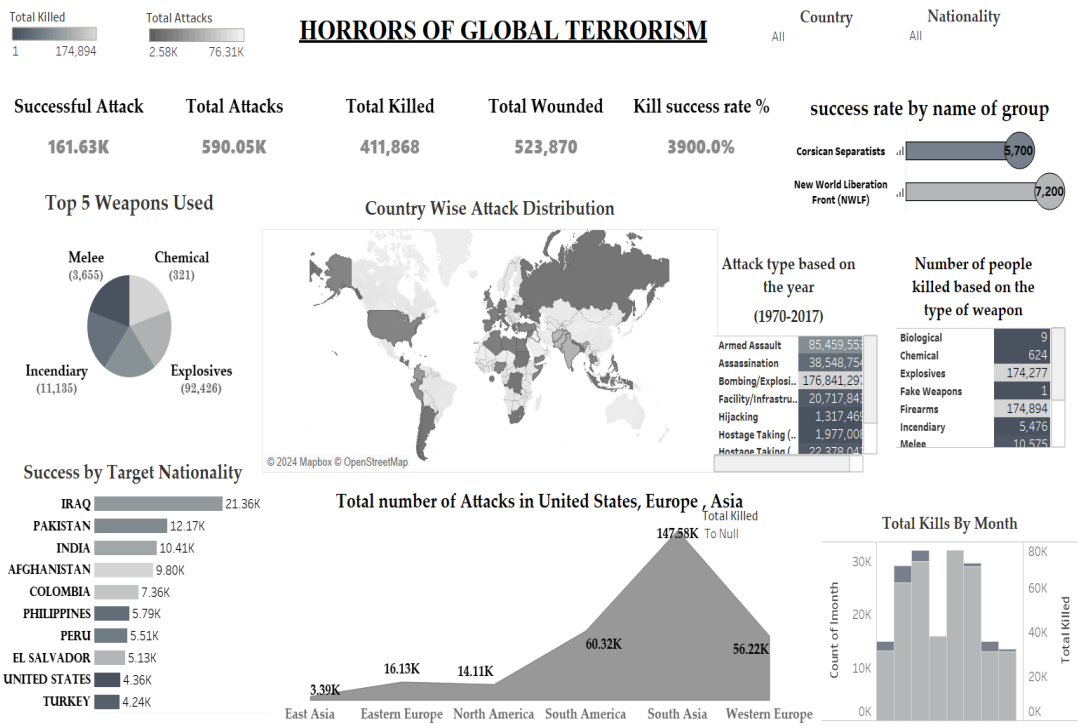
1. Attack type based on the year.
2. Types of weapons used.
3. Number of people killed based on the type of weapon.
4. Number of attacks by country
5. Types of attacks

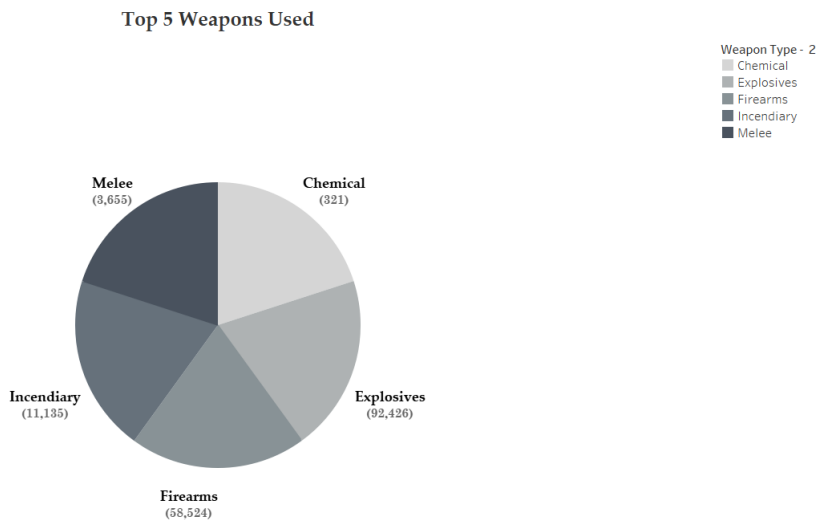
What is terrorism?

1. Terrorism is defined in the Oxford Dictionary as "the unlawful use of violence and intimidation, especially against civilians, in the pursuit of political aims." The key problem is that terrorism is difficult to distinguish from other forms of political violence and violent crime, such as state-based armed conflict, non-state conflict, one-sided violence, hate crime, and homicide.

2. In our coverage of terrorism, we rely strongly on data from the Global Terrorism Database (GTD), which defines terrorism as "acts of violence by non-state actors, perpetrated against civilian populations, intended to cause fear, in order to achieve a political objective."

DASHBOARD OF THE PROJECT

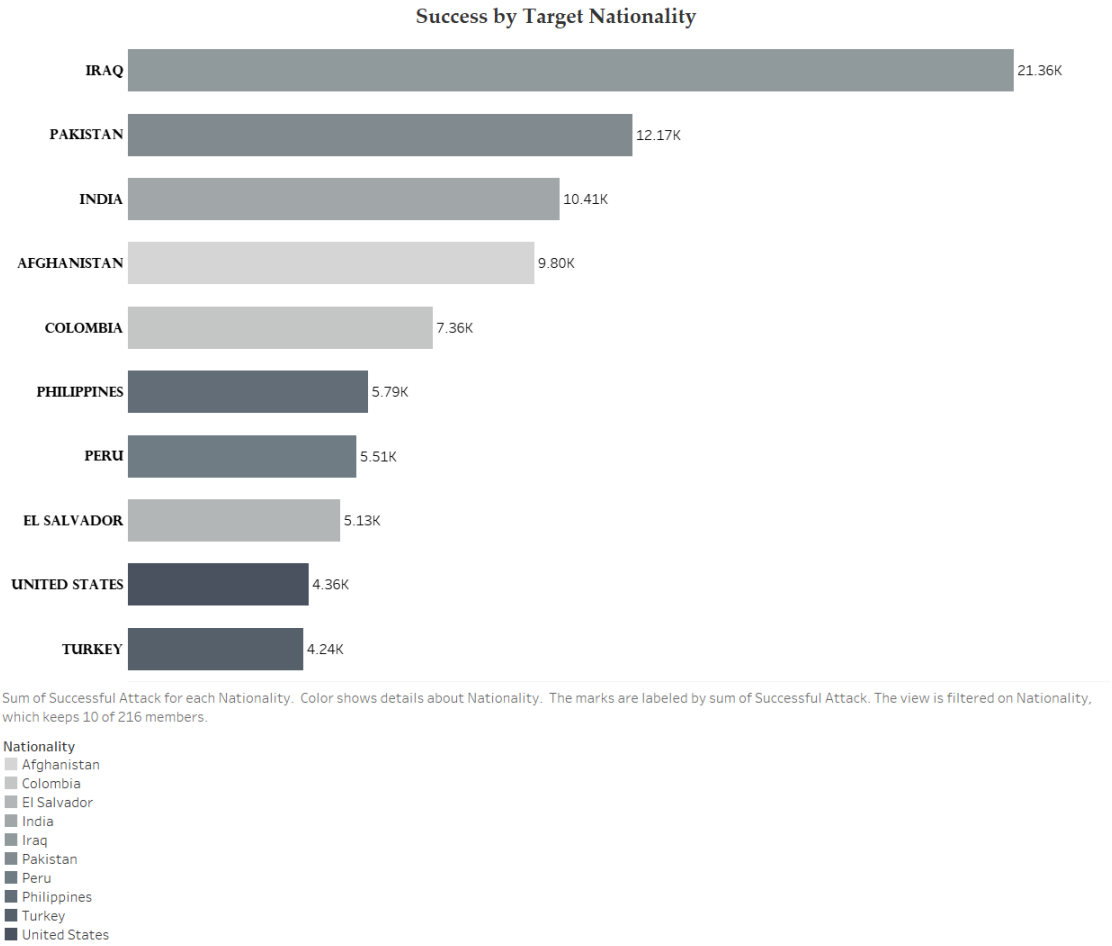


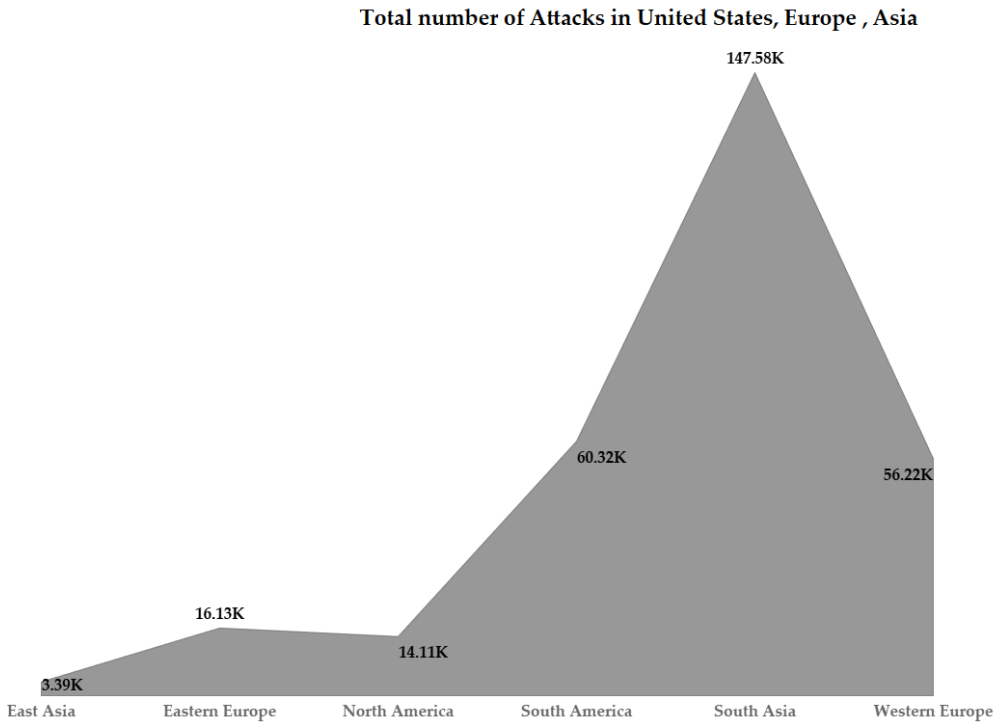


Count of Weapon Type - 2 and Weapon Type - 2. Color shows details about Weapon Type - 2. The marks are labeled by count of Weapon Type - 2 and Weapon Type - 2. The view is filtered on Weapon Type - 2, which keeps Chemical, Explosives, Firearms, Incendiary and Melee.

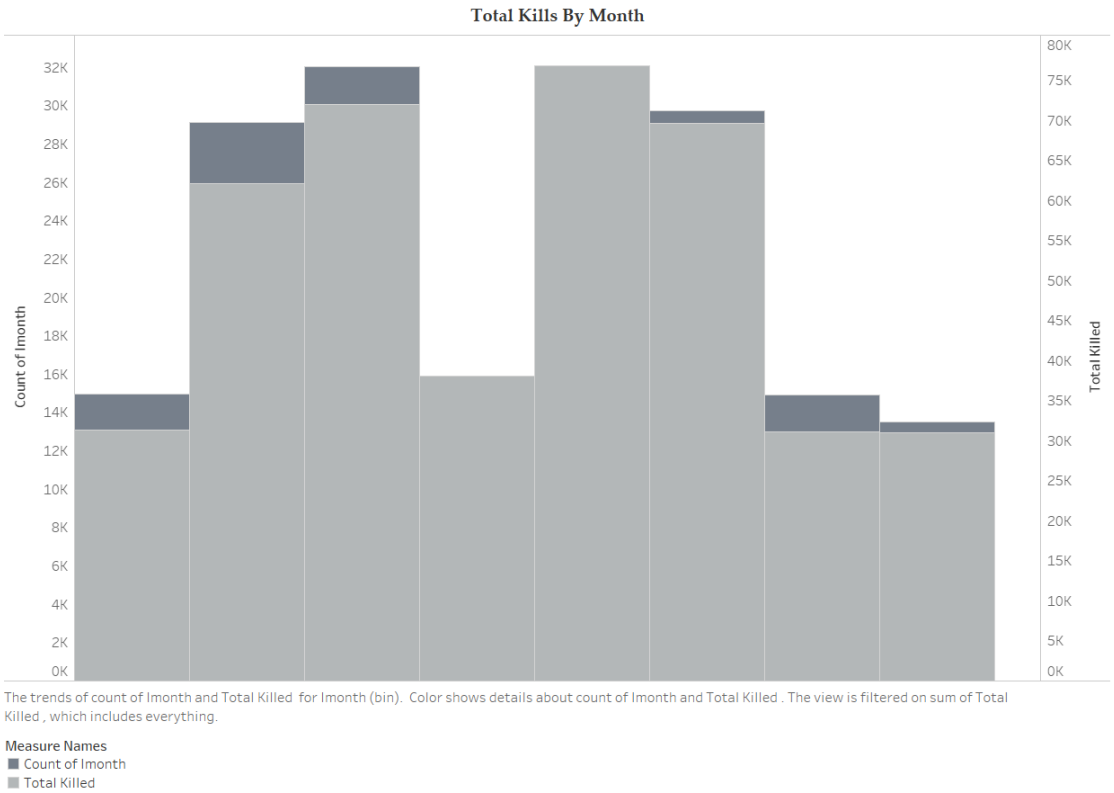
The pie chart illustrates the distribution of weapon usage, with the top 5 weapons being clearly depicted based on their frequency of use between the year 1970- 2017.

The bar chart highlights the success rates across different target nationalities, focusing on the top 10 performers in terms of achievement





The chart represents the total number of attacks in the USA, Asia, and Europe, highlighting the distribution of weapon usage across these regions.



The chart depicts the total number of kills per month.

Descriptive analysis/output

Descriptive is the term given to the analysis of data that helps describe, show, or summarize data in a meaningful way such that, for example, patterns might emerge from the data. Descriptive do not, however, allow us to make conclusions beyond the data we have analysed or reach conclusions regarding any hypotheses we might have made. They are simply a way to describe our data.

Descriptive statistics are very important because if we simply presented our raw data, it would be hard to visualize what the data was showing, especially if

there was a lot of it. Descriptive therefore enables us to present the data in a more meaningful way, which allows simpler interpretation of the data.

What Went Well:

1. Data collection was thorough and comprehensive.
2. Visualizations effectively conveyed key trends and patterns.
3. The project successfully raised awareness about global terrorism issues.

What Did NOT go Well:

1. Some data sources were incomplete or outdated, leading to gaps in analysis.
2. Certain visualizations were overly complex and difficult to interpret for some audience members.
3. Limited time constraints hindered deeper exploration of certain aspects of the data.

What Would you do Differently Next Time:

1. Use more reliable and up-to-date data sources to ensure accuracy.
2. Simplify visualizations to make them more accessible to a wider audience.
3. Allocate more time for in-depth analysis and exploration of specific trends or regions.